Synchronous tune measurements at an extreme condition*

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A low-α optics (synchronous frequency below 5 kHz) is developed to generate a few picoseconds and intense synchrotron radiation in far-infrared to THz regime by coherence nature.

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I. INTRODUCTION

Generating picosecond and subpicosecond electron bunches is increasing interest from the user community. Short bunchs radiate short photon pulses from X-ray to the THz frequency range. In the THz frequency range, we can obtain the high time resolution by short pulses of emission of the bunch which enables a user to study chemical reactions state changing or can be used to inspect nuclear motion in phonon excitation. There is another advantage of the short bunch, THz radiation. In that region, the bunch can emit Coherent Synchrotron Radiation(CSR) by a magnet, which means the THz region the bunch length is shorter than the wavelengths. To reduce bunch length for making THz radiation, the main parameter is momentum compaction factor α. The bunch length is related to $\sqrt{\alpha}$. The small bunch length can be used by groups of users who want to handle coherent synchrotron radiation.

To reach a low bunch length in quite short electron bunches, an isochronous operation in a synchrotron can be considered. The isochronous storage ring presents a small bunch length under the magnitude of α smaller than the conventional synchrotrons, which determines accelerator conditions ought to operate under low-α mode for both operating the isochronous operation and generating a few picosecond bunch lengths. However, small bunches mean the beam cannot store a high current of more than only a few microamperes, and a few picosecond bunch lengths make beam instability that stable beam operating is more difficult than normal α mode. Furthermore, under low-α conditions, accelerator synchronous frequency f_s is below 5 kHz. Various noise components, including atmospheric and man-made noise, dominate the region of a few kHz. This includes acoustic noise from cooling pipes (ranging from Hz to kHz), electrical signals from fast switches (ranging from kHz to MHz), and main power noise (~ 50 Hz), all of which dominate the thermal noise. For these reasons, the tune measurement with the old method is rather challenging to obtain enough signal amplitude. It means we cannot decide whether the beam status is stable or not, which means we cannot control the beam exact timing to maintain the beam stable, leading to raised beam instability. Nonetheless, because of the short bunches advantages, beamline users still prefer to utilize short bunches with the high average power and stability available by storage rings.

This paper proposed a different measurement method of synchronous tune with enough signal intensity to control the bunch exact timing. We considered that gauges the tune at the sideband of the betatron frequency via a coupling through chromaticity in a storage ring with inevitably exists in the machine to suppress beam instabilities. We confirmed that the tune can be measured at the sideband of the betatron motion through higher-order particle tracking simulation. Afterward, we designed an analog amplifier circuit to obtain enough beam signal and bandpass filters. Subsequently, we obtain the tune by the sideband of the betatron motion even under low-α mode, extreme conditions.

II. LONGITUDINAL BETATRON MOTION

The beam motion can be separated into longitudinal and transverse motions. To determine the amplitude of both longitudinal and transverse motions, we reexamined the equation of motion for equilibrium-state electrons in a storage ring. The bunches linear longitudinal dynamics, related to $(z, \delta = \Delta p/p)$, for the n-th rotation, can be expressed

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* A footnote to the article title

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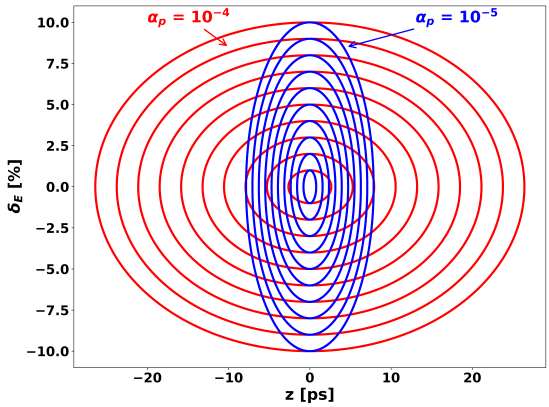

FIG. 1. Maximum oscillation amplitude in the longitudinal plane as a function of the initial energy spread and momentum compaction factor.

$$z(n) = z_0 \cos(2\pi\nu_s n) - \frac{\eta c}{\omega_s}\delta_0 \sin(2\pi\nu_s n),$$
$$\delta(n) = \frac{\omega_s}{\eta c} z_0 \sin(2\pi\nu_s n) + \delta_0 \cos(2\pi\nu_s n), \quad (1)$$

where z_0 and δ_0 are the initial coordinates in the longitudinal plane. The phase slip factor is denoted by $\eta = \alpha_p - 1/\gamma^2$, while γ is the Lorentz factor, α_p represents the first-order momentum compaction factor, and c is the speed of light. Oscillations offset of phase and amplitude within radio-frequency cavities deviate from the initial condition, introducing perturbations in energy, which leads to minor beam energy deviations without direct phase difference, i.e., $z_0 = 0$. When we incorporate the synchronous frequency $\omega_s^2 = -(eV_{RF}\omega_{RF}\alpha_p \cos\phi_s)/(E_0 T_0)$ primarily influenced by synchrotron radiation loss and the RF cavities energy contribution into Eq. 1, we can express the phase variations stemming from minor energy perturbations as $\Delta\delta$ can be written

$$z(n) = -\frac{\eta c\sqrt{E_0 T_0}}{\sqrt{-eV_{RF}\omega_{RF}\alpha_p \cos(\phi_s)}}\Delta\delta \sin(2\pi\nu_s n), \quad (2)$$

where V_{RF} and ω_{RF} are the effective voltage and frequency of an RF cavity, $\phi_s = \pi - sin(U_0/eV_{RF})$, U_0 is the energy loss per turn, E_0 denotes the kinetic energy of the beam, and T_0 is the revolution time. The oscillation amplitude of the phase shrinks when momentum compaction factor α_p is reduced as depicted in Eq. 2. In addition, the maximum oscillation amplitude in the longitudinal plane is calculated as a function of the initial energy spread and α_p with machine parameters listed in Table I.

The oscillation amplitude of the beam on the longitudinal plane shrinks when the α_p decreases. For this reason, the amplitude measurement on the longitudinal plane (or phase detection) can be challenging with low-α_p mode. Specifically, when operating low-α condition, both α_p and $\Delta\delta$ are $0.63 \times 10^{-4}(f_s \sim 5kHz)$ and 10^{-5}, respectively. These can be induced by phase and amplitude variations of RF fields. Typically, an RF-field generator system has a relative jitter of 10^{-4} in magnitude and 10^{-3} in phase, an order of magnitude higher under the low-level signal.

TABLE I. Machine parameters of Metrology Light Source

Circumference	48 m
Momentum compaction factor	1.3×10^{-4} to 3×10^{-2}
RF cavity voltage	500 kV
Beam current	200 mA
Bunch charge	400 pC
Injection energy	105 MeV
Operational energy	105 to 630 MeV
RF frequency	499.654 MHz
Revolution frequency	6.2457 MHz
harmonic number	80
horizontal emittance	120 to 200 nm rad
γ	1231
typical lifetime	4.5 to 10 h

In the phase detector perspective, the maximum amplitude in the longitudinal direction Δz_{max} is 5.96 $\mu m, \sim 20fs$, 10^{-5} @ 500 MHz, is quite a small value to detect the signal, which is a near lower limit of the cutting-edge phase detector based on RF signals. Hence, in the low-α mode, it is challenging to measure sufficient signal due to reduced bunch length.

III. TRANSVERSE MOTION BY AN ENERGY PERTURBATION

In a storage ring, the beam dynamics in a horizontal plane are dominated by synchrotron damping and quantum excitation. Once the beam is in an equilibrium state, the horizontal motion of the equation of the stored beams as a function of the number of turns is given by

$$x(n) = \sqrt{2\beta_x J_x} \cos\left[\psi_0 + \int_0^n 2\pi\nu_x ds\right], \quad (3)$$

where β_x is the betatron amplitude, J_x is the action, ν_x denotes the phase advanced per turn, and ψ_0 is the initial angle. A coupling between the horizontal and longitudinal planes is generated as a consequence of chromatic effects in accordance with the dispersion function. The phase advanced per turn can be described by three main phenomena e.g., pure betatron phase advance ν_{x0}, chromatic tune shift (with a horizontal chromaticity ξ_x), and amplitude-dependent tune shift, i.e., $\nu_x = \nu_{x0} + \xi_x\delta + \mu J$, where μ is the detuning factor and J is the total action. For our case, the phase term can be simplified as $\nu_x = \nu_{x0} + \xi_x\delta$ since the beam is in an equilibrium state (in order words, the offset is negligible), i.e., $J = 0$. By substituting Eq. (2) into Eq. (4) with an assumption

of $z_0 = 0$, the horizontal motion is described by using Jacobi-Anger expansion,

$$x(n) = \sqrt{2\beta_x J_x}\Re\left(e^{2\pi i\nu_{x0}n}e^{i\frac{\xi\delta_0}{\nu_s}\sin(2\pi\nu_s n)}e^{i\psi_0}\right), \quad (4)$$

$$= \sqrt{2\beta_x J_x}\sum_{k=-\infty}^{\infty} I_k(\frac{\xi_x\delta_0}{\nu_s})\cos(2\pi(\nu_{x0}+k\nu_s)n+\psi_0) \quad (5)$$

By taking the parameters that are used to estimate the longitudinal oscillation(α_p of 0.63×10^{-4} ($f_s \sim 5kHz$) and a small perturbation of $\Delta\delta = 10^{-5}$) as well as a chromaticity of 1 account for the maximum amplitude calculation at the betatron sideband, $\Delta x_{max,k=-1} = \Delta x_{max,k=1}$ is 8.63 μm which is far bigger than the capability of beam position monitors (BPMs) (in general, the resolution of BPMs is a few μm but it is relying on the input power which is defined by beam current (linearly proportional) and bunch length (changes the power in frequency domain)). Based on Eq. (6), the maximum oscillation amplitude in the horizontal plane is calculated as a function of the initial energy spread and momentum compaction factor.

IV. HIGH-ORDER ABERRATION

In the previous section, we only consider the linear dynamics of a stored beam. However, magnets practically derive nonlinear motions so-called high-order aberrations. This can lead to a catastrophic motion. The effect can be investigated by numerical simulations using Elegant code which offers 6-D macro-particle tracking simulations taking into account up to the 4th-order motion of equations with Lie Algebra through magnetic components such as bending, quadrupole, and sextupole magnets. The lattice functions used in the simulation are established based on linear optics from closed orbit measurements of the present Metrology Light Source (MLS). The momentum compaction factor has been controlled using two quadrupole magnets in a dispersive section. The baseline design has a momentum compaction factor of 10^{-4} and a horizontal chromaticity of about 1. This stimulates

A single particle tracking has been performed with an initial offset of 2×10^{-4} on energy spread direction.

In addition, synchrotron radiation plays an important role in all the 4-dimensional phase space, x-x' and z-δ_E, resulting in chaotic dilution of the betatron and synchronous motion due to the discrete energy variation of particles at each turn.

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Appendix A: Appendixes

Appendix B: A little more on appendixes

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